# (1) Array Creation Function

In [1]:
```
import numpy as np
```

In [2]:
```
# Create an array from a list
a = np.array([4, 6, 8])
print("Array a:", a)
```

```
Array a: [4 6 8]
```

In [3]:
```
# Create an array with evenly spaced values
b = np.arange(0, 15, 4) # Values from 0 to 10 with step 2
print("Array b:", b)
```

```
Array b: [ 0  4  8 12]
```

In [5]:
```
# Create an array filled with zeros
d = np.zeros((3, 4)) # 2x3 array of zeros
print("Array d:\n", d)
```

```
Array d:
 [[0. 0. 0. 0.]
 [0. 0. 0. 0.]
 [0. 0. 0. 0.]]
```

In [6]:
```
# Create an array filled with ones
e = np.ones((4, 2), dtype = int) # 3x2 array of ones
print("Array e:\n", e)
```

```
Array e:
 [[1 1]
 [1 1]
 [1 1]
 [1 1]]
```

In [7]:
```
#create an identity matrix

f = np.eye(4)   #4x4 identitymatrix
print("Identity matrix f:\n", f)
```

```
Identity matrix f:
 [[1. 0. 0. 0.]
 [0. 1. 0. 0.]
 [0. 0. 1. 0.]
 [0. 0. 0. 1.]]
```

# (2) Array Manipulation Function

In [8]:
```
# reshape an array

a1 = np.array([1,2,3])
```

```
reshaped = np.reshape(a1, (3,1))   #reshape to 1x3
print('Reshape array:', reshaped)
```

```
Reshape array: [[1]
 [2]
 [3]]
```

In [9]:
```
# Flatten an array

f1 = np.array([[1, 2, 3], [3, 4, 5], [5, 6, 7]])
flattened = np.ravel(f1)     #Flatten to 1d array
print("Flattened array:", flattened)
```

```
Flattened array: [1 2 3 3 4 5 5 6 7]
```

In [10]:
```
# Transpose an array

e1 = np.array([[1, 2], [3, 4]])
transposed = np.transpose(e1)   #Transpose the array
print("Transposed array:\n", transposed)
```

```
Transposed array:
 [[1 3]
 [2 4]]
```

In [12]:
```
# Stack arrays vertically

a2 = np.array([1, 2])
b2 = np.array([3, 4])
stacked = np.vstack([a2, b2])  # Stack a and b vertically
print("Stacked arrays:\n", stacked)
```

```
Stacked arrays:
 [[1 2]
 [3 4]]
```

In [13]:
```
# Stack arrays horizonatally

a2 = np.array([1, 2])
b2 = np.array([3, 4])
stacked = np.hstack([a2, b2]) # Stack a and b vertically
print("Stacked arrays:\n", stacked)
```

```
Stacked arrays:
 [1 2 3 4]
```

# (3) Mathematical Functions

In [15]:
```
# Add two arrays
g = np.array([1, 2, 3, 4])
added = np.add(g, 2) # Add 2 to each element
print("Added 2 to g:", added)
```

```
Added 2 to g: [3 4 5 6]
```

In [16]:
```
# Square each element
squared = np.power(g, 2) # Square each element
print("Squared g:", squared)
```

```
Squared g: [ 1  4  9 16]
```

In [17]:
```
# Square root of each element
sqrt_val = np.sqrt(g) # Square root of each element
print("Square root of g:", sqrt_val)
```

```
Square root of g: [1.         1.41421356 1.73205081 2.        ]
```

In [18]:
```
print(a1)
print(g)
```

```
[1 2 3]
[1 2 3 4]
```

In [19]:
```
print(a)
print(a1)
```

```
[4 6 8]
[1 2 3]
```

In [20]:
```
a3 = np.array([1, 2, 3])
dot_product = np.dot(a1, a) # Dot product of a and a1
print("Dot product of a1 and a:", dot_product)
```

```
Dot product of a1 and a: 40
```

In [21]:
```
print(a1)
print(g)
```

```
[1 2 3]
[1 2 3 4]
```

In [22]:
```
a3 = np.array([1, 2, 3])
dot_product = np.dot(a1, a) # Dot product of a and a1
print("Dot product of a1 and a:", dot_product)
```

```
Dot product of a1 and a: 40
```

In [23]:
```
print(a1)
print(g)
```

```
[1 2 3]
[1 2 3 4]
```

In [24]:
```
g
```

Out[24]:
```
array([1, 2, 3, 4])
```

In [25]:
```
a4 = np.array([1, 2, 3])
dot_product = np.dot(a1, g) # Dot product of a and a1
print("Dot product of a1 and a:", dot_product)
```

```
---------------------------------------------------------------------------
ValueError                                Traceback (most recent call last)
Cell In[25], line 2
      1 a4 = np.array([1, 2, 3])
----> 2 dot_product = np.dot(a1, g) # Dot product of a and a1
      3 print("Dot product of a1 and a:", dot_product)

ValueError: shapes (3,) and (4,) not aligned: 3 (dim 0) != 4 (dim 0)
```

# (4) Statistical Functions

In [26]:
```
s = np.array([1, 2, 3, 4])
mean = np.mean(s)
print("Mean of s:", mean)
```

```
Mean of s: 2.5
```

In [27]:
```
s = np.array([1, 2, 3, 4])
mean = np.var(s)
print("Mean of s:", mean) # variance
```

```
Mean of s: 1.25
```

In [28]:
```
# Standard deviation of an array
std_dev = np.std(s)
print("Standard deviation of s:", std_dev)
```

```
Standard deviation of s: 1.118033988749895
```

In [29]:
```
# Minimum element of an array
minimum = np.min(s)
print("Min of s:", minimum)
```

```
Min of s: 1
```

In [30]:
```
# Maximum element of an array
maximum = np.max(s)
print("Max of s:", maximum)
```

```
Max of s: 4
```

# (5) Linear Algebra Functions

In [32]:
```
# Create a matrix
matrix = np.array([[1, 2], [3, 4]])
```

In [33]:
```
matrix
```

Out[33]:
```
array([[1, 2],
       [3, 4]])
```

# (6) Random Sampling Functions

In [34]:
```
# Generate random values between 0 and 1
random_vals = np.random.rand(3) # Array of 3 random values between 0 and 1
print("Random values:", random_vals)
```

```
Random values: [0.63209174 0.85818386 0.12347586]
```

In [35]:
```
# Set seed for reproducibility
np.random.seed(0)
# Generate random values between 0 and 1
random_vals = np.random.rand(3) # Array of 3 random values between 0 and 1
print("Random values:", random_vals)
```

```
Random values: [0.5488135  0.71518937 0.60276338]
```

In [36]:
```
# Generate random integers
rand_ints = np.random.randint(0, 10, size=5) # Random integers between 0 and 10
print("Random integers:", rand_ints)
```

```
Random integers: [3 7 9 3 5]
```

In [37]:
```
# Set seed for reproducibility
np.random.seed(0)
# Generate random integers
rand_ints = np.random.randint(0, 10, size=5) # Random integers between 0 and 10
print("Random integers:", rand_ints)
```

```
Random integers: [5 0 3 3 7]
```

# (7) Boolean & Logical Functions

In [38]:
```
# Check if all elements are True
# all
logical_test = np.array([True, False, True])
all_true = np.all(logical_test) # Check if all are True
print("All elements True:", all_true)
```

```
All elements True: False
```

In [39]:
```
# Check if all elements are True
logical_test = np.array([True, False, True])
all_true = np.all(logical_test) # Check if all are True
print("All elements True:", all_true)
```

```
All elements True: False
```

In [40]:
```
# Check if all elements are True
logical_test = np.array([False, False, False])
all_true = np.all(logical_test) # Check if all are True
print("All elements True:", all_true)
```

```
All elements True: False
```

In [41]:
```
# Check if any elements are True
# any
```

```
any_true = np.any(logical_test) # Check if any are True
print("Any elements True:", any_true)
```

```
Any elements True: False
```

# (8) Set Operations

In [42]:
```
# Intersection of two arrays
set_a = np.array([1, 2, 3, 4])
set_b = np.array([3, 4, 5, 6])
intersection = np.intersect1d(set_a, set_b)
print("Intersection of a and b:", intersection)
```

```
Intersection of a and b: [3 4]
```

In [43]:
```
# Union of two arrays
union = np.union1d(set_a, set_b)
print("Union of a and b:", union)
```

```
Union of a and b: [1 2 3 4 5 6]
```

# (9) Array Attribute Functions

In [45]:
```
# Array attributes
a = np.array([1, 2, 3])
shape = a.shape # Shape of the array
size = a.size # Number of elements
dimensions = a.ndim # Number of dimensions
dtype = a.dtype # Data type of the array
print("Shape of a:", shape)
print("Size of a:", size)
print("Number of dimensions of a:", dimensions)
print("Data type of a:", dtype)
```

```
Shape of a: (3,)
Size of a: 3
Number of dimensions of a: 1
Data type of a: int64
```

# (10) Other Functions

In [46]:
```
# Create a copy of an array
a = np.array([1, 2, 3])
copied_array = np.copy(a) # Create a copy of array a
print("Copied array:", copied_array)
```

```
Copied array: [1 2 3]
```

In [47]:
```
# Size in bytes of an array
array_size_in_bytes = a.nbytes # Size in bytes
print("Size of a in bytes:", array_size_in_bytes)
```

```
Size of a in bytes: 24
```

In [48]:
```
# Check if two arrays share memory
shared = np.shares_memory(a, copied_array) # Check if arrays share memory
print("Do a and copied_array share memory?", shared)
```

```
Do a and copied_array share memory? False
```

In [ ]: